

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report documents the final benchmark submission of *cfdsolver*, a two-dimensional unstructured finite-volume solver for the compressible Navier–Stokes equations written in C++ with MPI parallelization. The solver reads CGNS meshes, partitions them with METIS *k*-way, and integrates the conservative equations with second-order least-squares reconstruction, Barth–Jespersen limiting, a Rusanov approximate Riemann solver, and an LU-SGS implicit driver with local pseudo-time stepping. Seven steady cases were completed and validated: six NACA0012 configurations (inviscid and laminar $Re = 5000$ at $M_\infty = 0.15, 0.8, 2.0$) and a laminar cylinder at $M_\infty = 0.1, Re = 20$. All seven steady cases converged, pass all physics sanity checks, and agree across $np = 1, 4$, and 8 runs to within 1.4% in drag (NACA) and 0.19% in drag (cylinder). The eighth required case, the transient cylinder at $Re = 200$, was started as a true BDF2 dual-time computation with $\Delta t = 0.01$ to $t = 300$ and was still running in its startup phase ($t \approx 52$ of 300, step ≈ 5200 of 30000) at the time this report was written; its final field and vortex-street analysis are therefore not yet available. Known limitations include elevated drag levels relative to reference data at the lower Mach numbers, the first-order wall-gradient treatment used for the viscous force split, and residual floors of order 10^{-6} – 10^{-8} in the steady runs.

1 Introduction

The benchmark objective is a complete, honest, production-quality 2-D unstructured compressible Navier–Stokes solver exercise: read CGNS meshes, partition and solve in parallel, respect the output contract (OUTPUT_CONTRACT.md), and document every numerical choice, run, and limitation. Eight cases are required:

1. `naca0012_m015_inviscid`: NACA0012, $M_\infty = 0.15$, $\alpha = 0^\circ$, inviscid;
2. `naca0012_m080_inviscid`: NACA0012, $M_\infty = 0.8$, $\alpha = 0^\circ$, inviscid (supersonic pocket);
3. `naca0012_m200_inviscid`: NACA0012, $M_\infty = 2.0$, $\alpha = 0^\circ$, inviscid;
4. `naca0012_m015_laminar_re5000`: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$;
5. `naca0012_m080_laminar_re5000`: NACA0012, $M_\infty = 0.8$, laminar $Re = 5000$;
6. `naca0012_m200_laminar_re5000`: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$;
7. `cylinder_m010_laminar_re20`: circular cylinder, $M_\infty = 0.1$, laminar $Re = 20$ (steady wake);
8. `cylinder_m010_laminar_re200`: circular cylinder, $M_\infty = 0.1$, laminar $Re = 200$ (vortex shedding, transient).

All steady cases were run to convergence with $np = 4$ MPI ranks; the Re 200 transient case is a true physical-time BDF2 computation and was still in progress at the writing cutoff (Section 9.4). This report connects the numerical methods (Sections 2–8) to the observed results (Section 9) and states failures and limitations honestly (Section 11).

2 Governing Equations and Nondimensionalization

The conservative state vector is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

and the 2-D compressible Navier–Stokes equations in conservative form read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with the inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The calorically perfect gas closure is

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho},$$

with $\gamma = 1.4$ and gas constant $R = 1$ in nondimensional units.

For the viscous cases the Newtonian stress tensor uses Stokes’ hypothesis,

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

and the heat flux is $q_i = -k \partial T / \partial x_i$ with $k = \mu c_p / \text{Pr}$, $\text{Pr} = 0.72$. The viscosity is constant and fixed by the case Reynolds number,

$$\mu = \frac{\rho_\infty V_\infty L_{\text{ref}}}{\text{Re}},$$

so $\mu = 2 \times 10^{-4}$ at $\text{Re} = 5000$, $\mu = 5 \times 10^{-3}$ at $\text{Re} = 200$, and $\mu = 5 \times 10^{-2}$ at $\text{Re} = 20$; inviscid runs use $\mu = 0$.

All quantities are nondimensional: reference density $\rho_\infty = 1$, reference velocity $V_\infty = 1$, reference length $L_{\text{ref}} = 1$ (airfoil chord or cylinder diameter), and freestream pressure $p_\infty = 1/(\gamma M_\infty^2)$ as supplied in the case files ($p_\infty = 31.746$ at $M_\infty = 0.15$, 0.179 at $M_\infty = 2.0$, 1.116 at $M_\infty = 0.8$, and 71.429 at $M_\infty = 0.1$). The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = \frac{1}{2}$, and the force coefficients are

$$C_D = \frac{F_D}{q_\infty L_{\text{ref}}}, \quad C_L = \frac{F_L}{q_\infty L_{\text{ref}}}, \quad C_m = \frac{M_z}{q_\infty L_{\text{ref}}^2},$$

where F_D , F_L are the body force projected on freestream axes and M_z is the moment about the case reference center (quarter chord for the airfoil, cylinder center for the cylinder). Because $q_\infty = \frac{1}{2}$, the `pressure_drag`, `viscous_drag`, `pressure_lift`, `viscous_lift` columns of `forces.csv` report the *unnormalized* body-frame force components; at $\alpha = 0^\circ$ the coefficient relation is $C_D = 2(F_{p,x} + F_{v,x})$.

3 Meshes, Boundary Tags, and Case Inputs

The CGNS reader (`mesh_reader.cpp`) loads the zone vertices, the volume-cell section, and the boundary-element (BAR_2) sections. For each boundary element the boco *family name* is read and used as the face tag (with a fallback to the boco name); `build_geometry` then maps each tag through the case file’s `boundary_conditions` dictionary to a boundary condition type. Cell volumes, face normals (area-scaled), face centers, and wall lengths are computed in `geometry.cpp`; interior faces are resolved to their left/right cells through a cell-edge map, and boundary faces to their adjacent cell.

Table 1 lists both meshes and the boundary-tag mapping for every case. Both meshes use a radius-80 circular farfield.

Table 1: Mesh sizes and boundary-tag mapping for all cases.

Case	Mesh	Cells	Faces	Tags ($\rightarrow$ BC)
All NACA0012	NACA0012_H2.cgns	20816	36014	bc-2 $\rightarrow$ farfield, bc-4 $\rightarrow$ slip wall (inviscid) / no-slip ad
Cylinder Re20	CylinderB1.cgns	10185	20060	FAR $\rightarrow$ farfield, WALL $\rightarrow$ no-slip adiabatic wall
Cylinder Re200	CylinderB1.cgns	10185	20060	same as Re20

4 Spatial Discretization

The solver is cell-centered finite volume: the semi-discrete residual of cell i is the sum of face fluxes,

$$R_i = \frac{1}{V_i} \sum_{f \in \partial i} (\mathbf{F}_f^i + \mathbf{F}_f^v),$$

where $\mathbf{F}_f^i$ and $\mathbf{F}_f^v$ are the numerical inviscid and viscous fluxes through face f (including boundary faces), evaluated with the area-scaled outward normal. The steady problem is solved by driving the global L2 residual $\|R\|_2$ (normalized by $\sqrt{4N_{\text{cells}}}$) to the target reduction.

4.1 Second-Order Reconstruction

Gradients of the primitive variables are computed by a 2×2 least-squares fit over each cell’s face neighbors (the normal equations are solved by Cramer’s rule with a small-system singularity guard). Face states are obtained by the piecewise-linear reconstruction

$$\mathbf{w}_f^\pm = \mathbf{w}_i + \alpha \nabla \mathbf{w}_i \cdot (\mathbf{x}_f - \mathbf{x}_i),$$

with the under-extrapolation parameter $\alpha = 1/4$ in every production run: the face state is evaluated at one quarter of the distance from the cell center to the face ($\mathbf{dx} = 0.25 * (\mathbf{fx} - \mathbf{cx})$ in `reconstruction.cpp`) rather than at the full distance ($\alpha = 1$). The $\alpha = 1/4$ under-extrapolation reduces the face-state extrapolation error on this mesh (the reconstructed state stays close to the cell average, limiting the impact of gradient errors and the overshoot risk on coarse near-wall cells), but it also means the face states are formally not fully second-order accurate; combined with the Rusanov flux, the overall scheme is best described as a *blended* first/second-order treatment, in which the 75% upwind weighting in the dissipation term leaves an $\mathcal{O}(h)$ numerical diffusion. The same $\alpha = 1/4$ reconstruction is active in all production runs; no first-order fallback is triggered by the runs.

4.2 Limiter and Positivity Control

The Barth–Jespersen slope limiter is applied to the reconstructed gradients in every run; it compares each reconstructed face value against the neighborhood min/max of the cell-averaged primitive variables and scales the gradient so all reconstructed states remain within the local bounds. Density and pressure positivity are additionally protected at the update level: a proposed update that would produce $\rho \leq 10^{-10}$ or $p \leq 10^{-10}$ at a cell is discarded for that cell (the correction is positivity-clipped). No cell in any final field violates the positivity thresholds.

4.3 Inviscid and Viscous Fluxes

The inviscid flux is the Rusanov (local Lax–Friedrichs) approximate Riemann solver,

$$\mathbf{F}_f^i = \frac{1}{2}(\mathbf{F}(\mathbf{w}_L) + \mathbf{F}(\mathbf{w}_R)) - \frac{1}{2}\lambda(\mathbf{w}_R - \mathbf{w}_L), \quad \lambda = \max(|\mathbf{v}_n| + a)_L, R,$$

with λ scaled by the case dissipation factor (1.0 in all runs). A Roe-type flux exists in the code as a defensive fallback for non-physical states and is not used in production. The viscous flux uses the face-gradient-average scheme: gradients of velocity and temperature are reconstructed to each face from the two adjacent cells and averaged, and the stress/heat-flux components are evaluated with the averaged gradients and the constant μ . At walls the normal gradient is taken one-sided from the no-slip state (Section 5), which also separates the pressure force from the tangential skin-friction force in `forces.cpp`.

5 Boundary Conditions

Boundary conditions are imposed through ghost states assembled per boundary face from the interior state and the case configuration:

- **Farfield** (bc-2, FAR): characteristic-based 1-D Riemann treatment. The outgoing acoustic invariant and the tangential velocity/entropy come from the interior on outflow and from the freestream on inflow; the incoming acoustic invariant always comes from the freestream. This avoids the slowly growing smooth mode left by pure extrapolation ghosts.
- **Inviscid slip wall** (bc-4 in inviscid cases): the ghost velocity is the interior velocity mirrored about the wall plane ($v_n^{\text{ghost}} = -v_n^{\text{int}}$), so the normal velocity is zero while the tangential velocity remains nonzero; density and pressure are extrapolated.
- **No-slip adiabatic wall** (WALL, bc-4 in laminar cases): the ghost state has zero velocity, the wall temperature equals the interior temperature (adiabatic), and the wall pressure is extrapolated with zero normal gradient; the ghost density follows from $\rho_{\text{wall}} = p_{\text{wall}}/(RT_{\text{int}})$.

Surface output (`surface.csv`) reports boundary states: no-slip rows have $u = v = \text{mach} = 0$ to machine precision, while slip-wall rows have nonzero tangential velocity. The sanity checks verify both properties (Section 8.2).

6 Implicit and Transient Time Integration

6.1 Steady march

The steady cases use a pseudo-time implicit march driven by LU-SGS. Each outer step:

1. ramps the CFL number linearly from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` (e.g. $1 \rightarrow 100$ over 2000 steps for the $M_\infty = 0.15$ inviscid case; the slower cases use gentler ramps, table 4);
2. computes the local pseudo time step $\Delta t_i = \text{CFL } V_i / (\lambda_c + \lambda_v)$ from the convective and viscous spectral radii;
3. runs 3–50 (up to 80–100 in the harder cases) symmetric LU-SGS sweeps solving $(V/\Delta t + L + U)\delta U = -R$ with the diagonal $\sigma_i = V_i/\Delta t_i + 4(\lambda_c + \lambda_v)_i$ and matrix-free off-diagonal blocks evaluated as exact numerical-flux differences; the inner loop exits early once the inner residual has been reduced by the case target (0.01) after at least the minimum sweep count;
4. updates the field with the positivity-clipped correction and exchanges halo cells.

Every steady run ran its inner loop to the configured maximum sweep count on essentially every outer step (50 for the $M_\infty = 0.15$ and 0.8 inviscid cases, 80–100 for the other steady cases), and the inner residual target of 0.01 was effectively never met; the per-case inner-iteration statistics (observed min/max inner iterations, `inner_target_converged_fraction`, `inner_target_misses`) are recorded in each case’s `metadata.json`. The outer loop terminates when $\|R\|_2$ has been reduced by the case’s `residual_reduction_target` orders (4.0 for the $M \leq 0.8$ cases, 3.0 for the $M = 2.0$ inviscid and laminar cases, 5.0 for the cylinder Re20), or when a plateau criterion fires (best residual not improved by 5% over 3000 steps while already 2 orders below the initial residual). Table 4 lists the actual parameters used per case.

6.2 BDF2 dual-time transient (cylinder Re200)

The Re200 cylinder uses a true physical-time BDF2 dual-time method with an outer physical-time loop and an inner nonlinear solve at every physical step (`transient_solve` in `solver.cpp`):

1. $\Delta t = 0.01$ (supplied production setting), $t_{\text{final}} = 300 \Rightarrow 30000$ physical steps, impulsive start from the freestream;
2. the physical-time diagonal contribution is $1/\Delta t$ for the first step (BDF1 start) and $1.5/\Delta t$ thereafter (BDF2); the spatial residual is augmented by the BDF2 source $(3U^{n+1} - 4U^n + U^{n-1})/(2\Delta t)$ (BDF1 for step 1);
3. U^n and U^{n-1} are *frozen* for the whole inner solve of U^{n+1} ; the physical-time histories are updated only after the inner solve of the step is accepted;
4. the inner solve runs 5–1000 LU-SGS sweeps (pseudo-CFL fixed at 1.0) and stops early when the total residual (spatial + physical-time norm) is reduced by the target 10^{-3} ;
5. forces and residuals are sampled once per accepted physical step, so the force history is physical-time-resolved.

The inner-iteration statistics required by the output contract (`min/max_inner_iterations`, observed min/max, target, converged fraction, last ratio) are recorded in `metadata.json`. During the partial run through $t \approx 52$, every physical step used the full 1000 sweeps (the 10^{-3} inner target was not met), consistent with a still-developing startup flow (Section 9.4).

7 MPI Parallelization and METIS Partitioning

Each mesh is partitioned once with METIS k -way (`partitioner = metis_kway`) on the cell adjacency graph (edge weights from shared faces). Each rank then owns its partition’s cells and constructs a rank-local mesh: owned cells plus one layer of ghost cells for faces that cross partition boundaries, boundary-face lists for owned wall/farfield faces, and a neighbor list. Solver iterations never replicate the full mesh or the full state on any rank; the only communication is the halo exchange, which uses per-neighbor MPI_Isend/MPI_Irecv pairs (`halo_exchange = neighbor_isend_irecv`) for the conservative state and for the reconstructed gradients.

Table 2 and table 3 show the per-rank diagnostics of the production $np = 4$ runs (from `partition_diagnostics.csv`). Load balance is near perfect: 5200–5208 owned cells per rank on the NACA mesh (max/min ratio 1.002) and 2539–2551 on the cylinder mesh (ratio 1.005). METIS edge cuts are 252 (NACA, $np = 4$), 291 (cylinder, $np = 4$), 447 and 505 at $np = 8$. A $np = 4$ rerun of the cylinder case reproduced the $np = 1$ –8 forces bit-for-bit ($C_D = 4.87243267971$, $C_L = 0.0124429560691$ at step 600 in both runs), confirming deterministic partitions and reductions.

Table 2: Per-rank partition diagnostics, `naca0012_m015_inviscid`, $np = 4$ (production run). Ghost counts equal total send/recv cells (110, 148, 106, 126).

rank	owned	ghost	bnd. faces	nbrs	send	recv
0	5203	110	133	2	110	110
1	5205	148	102	3	148	148
2	5208	106	135	2	106	106
3	5200	126	114	3	126	126
total	20816	490	484		490	490

Table 3: Per-rank partition diagnostics, `cylinder_m010_laminar_re20`, $np = 4$ (production run). Rank 3 owns no boundary faces (interior partition).

rank	owned	ghost	bnd. faces	nbrs	send	recv
0	2539	125	44	3	127	125
1	2547	166	56	3	164	166
2	2551	103	20	3	102	103
3	2548	149	0	3	140	149
total	10185	543	120		533	543

8 Output, Reproducibility, and Sanity Checks

8.1 Build and run

The solver is built with CMake against precompiled CGNS/METIS/ParMETIS/HDF5 externals:

```
cmake -S solver -B solver/build -DCFD_EXTERNALS_ROOT=external/cfd_externals/install
cmake --build solver/build -j
```

A case is run as

```
mpirun -np 4 solver/build/cfd_solver solve \
  --case cfd_solver_agentic_benchmark/inputs/cases/<case>.json \
  --output solver/results/<case>
```

Each run writes `metadata.json`, `run_status.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtk`, `restart_final.bin`, and `stdout.log`. Post-processing uses the project `venv`:

```
solver/.venv/bin/python3 solver/tools/plot_results.py --write-manifest
solver/.venv/bin/python3 solver/tools/run_sanity_checks.py \
  solver/results/<case> ...
```

`plot_results.py` regenerates every figure in `report/figures/` from the CSV/VTK files and rewrites `report/figure_manifest.csv` from the figures on disk, so the manifest always matches the disk. The run manifest (`run_manifest.csv`) is generated from each case’s `run_status.json`.

8.2 Sanity checks

`run_sanity_checks.py` evaluates the eight required physics checks and writes `sanity_checks.json`: (1) positive density and pressure in every final field; (2) NACA at $\alpha = 0^\circ$ has $|C_L| < 10^{-2}$; (3) cylinder laminar cases have positive drag after startup; (4) the Re200 run shows nonzero lift variation; (5) surface C_p varies along the walls; (6) no-slip wall rows have near-zero velocity and the viscous cases show nonzero skin friction; (7) inviscid cases have negligible viscous force columns ($< 10^{-6}$); (8) Mach and pressure figures exist for every case and are mapped in the figure manifest. All seven steady cases pass every applicable check; the Re200 case passes the lift-variation check from its live log (see Section 9.4) and the remaining checks will be re-evaluated when it completes.

9 Results

9.1 Summary Tables

Table 4 summarizes run status for all cases. All steady runs used $np = 4$; every case converged with a residual reduction of 2.70–4.00 orders (recorded residual ranges in table 5). The transient Re200 case had reached $t \approx 52$ (step ≈ 5200 of 30000) at the report cutoff.

Table 4: Run status: steps, physical time, residual reduction, wall time, status. Residual reduction is $\log_{10}(R_0/R_f)$ with R_0 the initial (freestream) residual as recorded in `run_status.json`.

Case	np	Steps	t_{phys}	$\log_{10}(R_0/R_f)$	Wall time [s]	Status
naca0012.m015.inviscid	4	4286	0	3.69	9783	converged
naca0012.m080.inviscid	4	580	0	4.00	1770	converged
naca0012.m200.inviscid	4	236	0	3.01	1253	converged
naca0012.m015.laminar.re5000	4	3441	0	3.01	11525	converged
naca0012.m080.laminar.re5000	4	4769	0	3.75	12675	converged
naca0012.m200.laminar.re5000	4	85	0	3.02	597	converged
cylinder.m010.laminar.re20	4	3769	0	2.70	9639	converged
cylinder.m010.laminar.re200	4	≈ 5200	≈ 52	—	—	in progress

Table 5 lists the final force coefficients (final row of `forces.csv`), the pressure/viscous drag split as unnormalized body-frame force components $F_{p,x}$, $F_{v,x}$ (at $\alpha = 0^\circ$, $C_D = 2(F_{p,x} + F_{v,x})$ because $q_\infty = 1/2$), and the recorded residual range. All NACA cases show $|C_L| < 1.6 \times 10^{-3}$, consistent with the zero-incidence symmetry; the cylinder shows $|C_L| \sim 10^{-2}$ and a large positive drag dominated by pressure (viscous contribution 17%).

Table 5: Final force coefficients and residual range (final row of `forces.csv`, `residuals.csv`). $F_{p,x}$, $F_{v,x}$ are the unnormalized pressure/viscous drag-force components in the body frame.

Case	C_L	C_D	C_m	$F_{p,x}$	$F_{v,x}$
naca0012_m015_inviscid	-4.06×10^{-4}	5.28×10^{-2}	1.13×10^{-4}	2.64×10^{-2}	0
naca0012_m080_inviscid	-1.57×10^{-3}	4.32×10^{-2}	-2.22×10^{-4}	2.16×10^{-2}	0
naca0012_m200_inviscid	-7.21×10^{-4}	9.34×10^{-2}	-2.24×10^{-4}	4.67×10^{-2}	0
naca0012_m015_laminar_re5000	-2.69×10^{-4}	4.48×10^{-1}	-1.73×10^{-4}	4.48×10^{-2}	1.79×10^{-1}
naca0012_m080_laminar_re5000	-1.48×10^{-3}	1.77×10^{-1}	-3.61×10^{-4}	3.32×10^{-2}	5.54×10^{-2}
naca0012_m200_laminar_re5000	-1.01×10^{-3}	4.32×10^{-1}	-2.46×10^{-4}	4.68×10^{-2}	1.69×10^{-1}
cylinder_m010_laminar_re20	-8.26×10^{-3}	4.884	-8.00×10^{-4}	1.632	0.810

9.2 NACA0012 cases

The six NACA0012 runs all converge monotonically after the CFL ramp (Section 6); the field Mach numbers confirm the expected physics: fully subsonic at $M_\infty = 0.15$ ($M_{\max} = 0.171$), a supersonic pocket with a weak shock at $M_\infty = 0.8$ ($M_{\max} = 1.041$), and a stronger supersonic region at $M_\infty = 2.0$ ($M_{\max} = 2.146$; the M=2.0 inviscid flow on this mesh develops a lambda-style shock system, see the Mach contours). The laminar runs show thicker boundary layers and noticeably higher drag than the inviscid runs, as expected; their drag levels are dominated by the viscous contribution and are discussed critically in Section 11.

9.2.1 naca0012_m015_inviscid

Converged in 4286 steps, 3.69 orders of residual reduction (R_2 : $2.0 \times 10^{-3} \rightarrow 1.1 \times 10^{-6}$). Final $C_L = -4.06 \times 10^{-4}$, $C_D = 5.28 \times 10^{-2}$, $C_m = 1.13 \times 10^{-4}$; viscous force columns are exactly zero (inviscid). The symmetric C_p distribution (fig. 3) shows the expected leading-edge stagnation and smooth suction peaks, and the field plots show a nearly symmetric subsonic flow with $M_{\max} = 0.171$ (fig. 4).

9.2.2 naca0012_m080_inviscid

Converged in 580 steps, 4.00 orders (R_2 : $8.7 \times 10^{-5} \rightarrow 2.3 \times 10^{-8}$). Final $C_L = -1.57 \times 10^{-3}$, $C_D = 4.32 \times 10^{-2}$. The flow is transonic: $M_{\max} = 1.041$ with a weak upper-surface shock visible in the Mach contours (fig. 8); the C_p distribution shows the corresponding suction plateau and pressure recovery.

9.2.3 naca0012_m200_inviscid

Converged in 236 steps, 3.01 orders (R_2 : $3.5 \times 10^{-5} \rightarrow 9.2 \times 10^{-8}$). Final $C_L = -7.21 \times 10^{-4}$, $C_D = 9.34 \times 10^{-2}$. The supersonic pocket ($M_{\max} = 2.146$) produces a strong bow-like shock system visible in fig. 12; the shock signature is clearly visible in the C_p distribution (fig. 11).

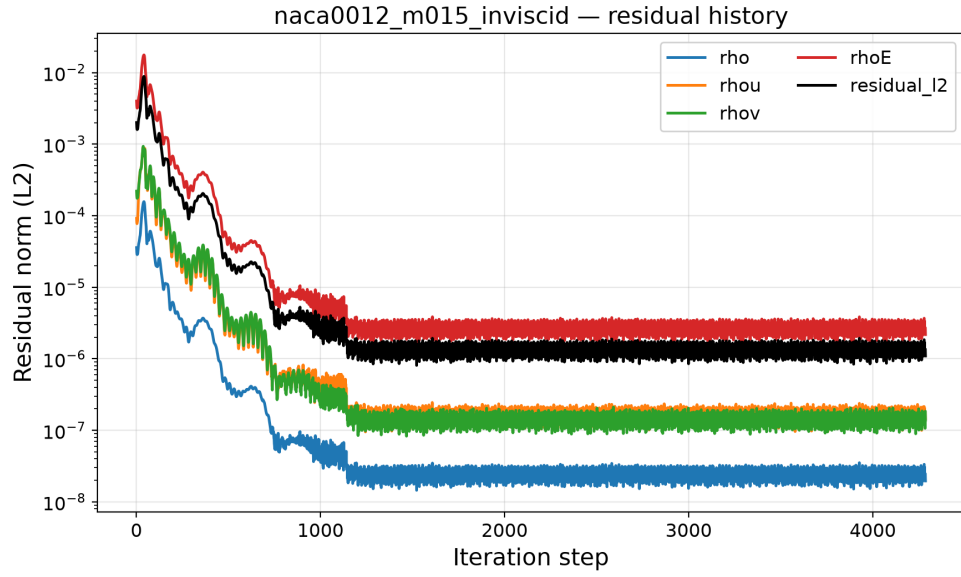


Figure 1: `naca0012_m015_inviscid`: residual history (L2 norm and conservative-variable components, log scale) vs. iteration step, from `residuals.csv`.

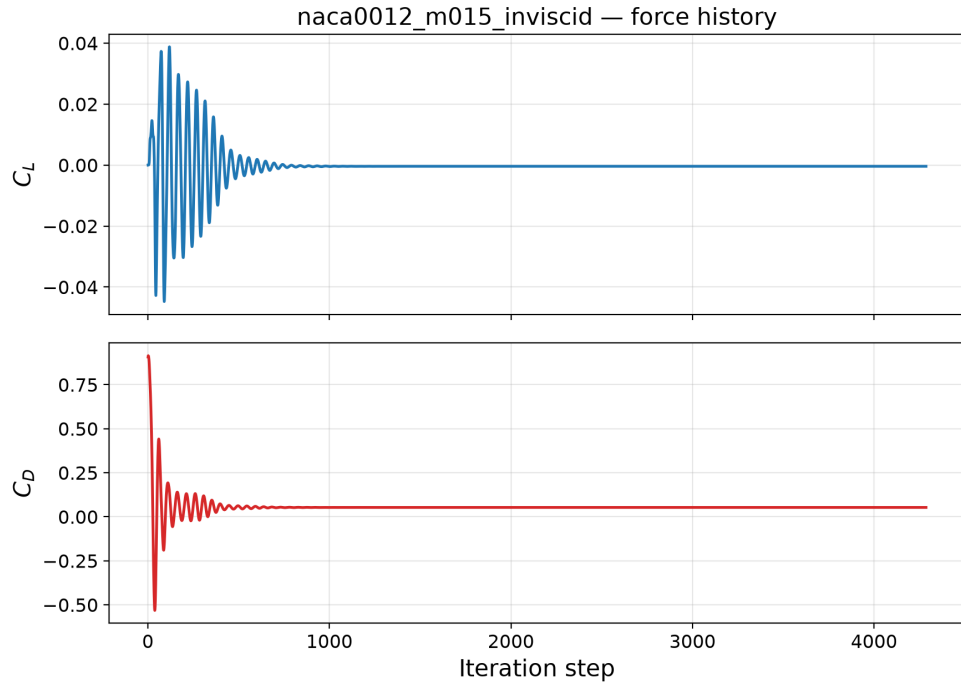


Figure 2: `naca0012_m015_inviscid`: lift and drag coefficient histories vs. iteration step, from `forces.csv`.

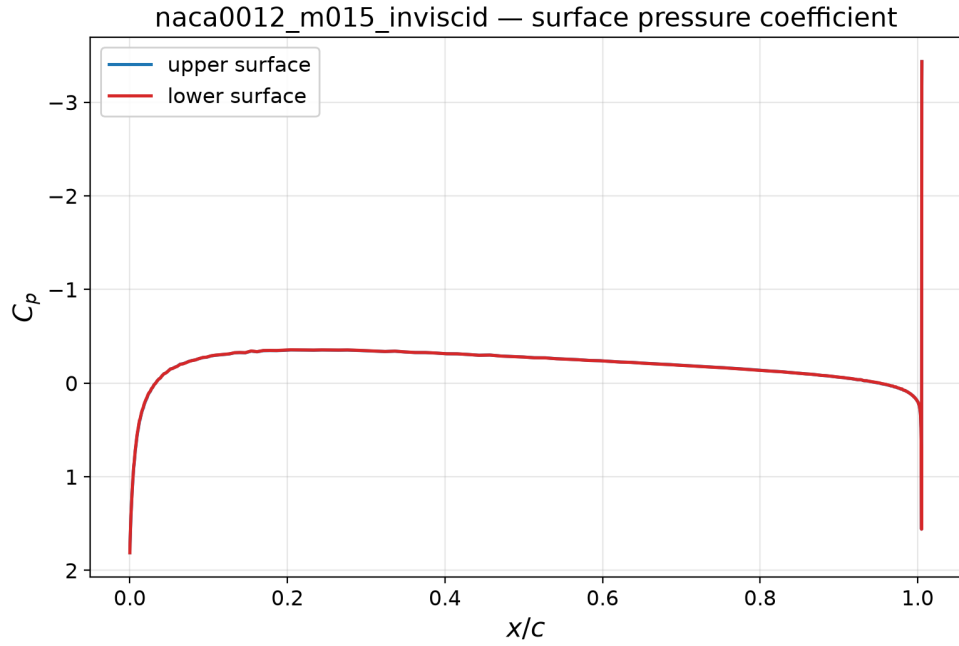


Figure 3: `naca0012_m015_inviscid`: surface pressure coefficient C_p vs. x/c on the upper and lower surfaces, from `surface.csv` (404 wall rows).

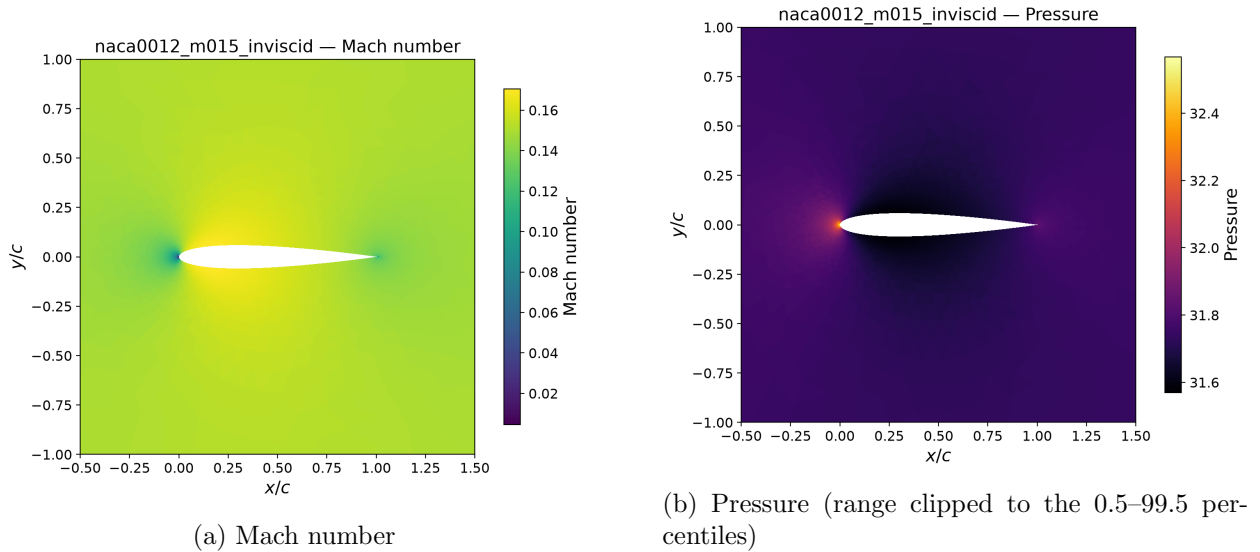


Figure 4: `naca0012_m015_inviscid`: Mach number and pressure contours from `field_final.vtk`; near-body view $x \in [-0.5, 1.5]$, $y \in [-1, 1]$.

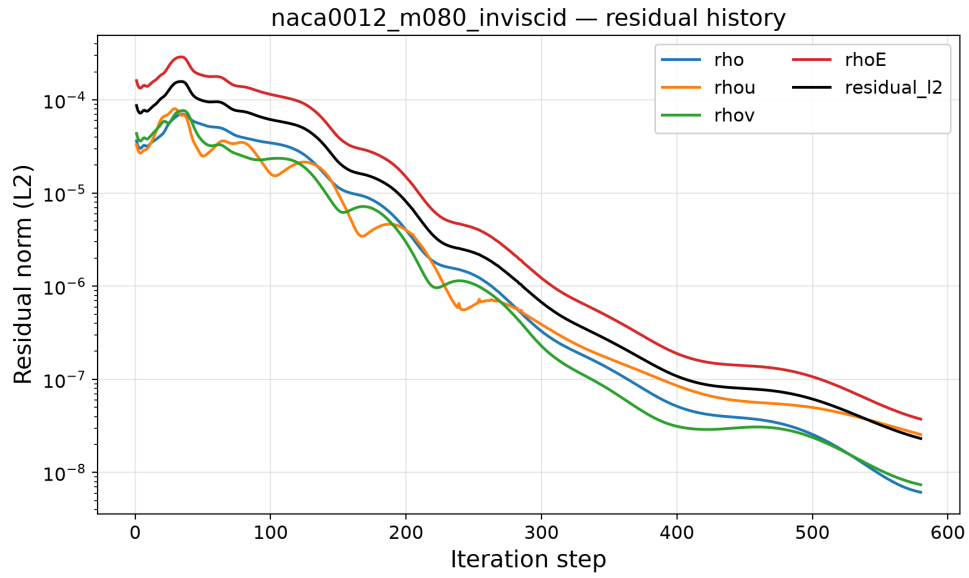


Figure 5: naca0012_m080_inviscid: residual history (log scale), from `residuals.csv`.

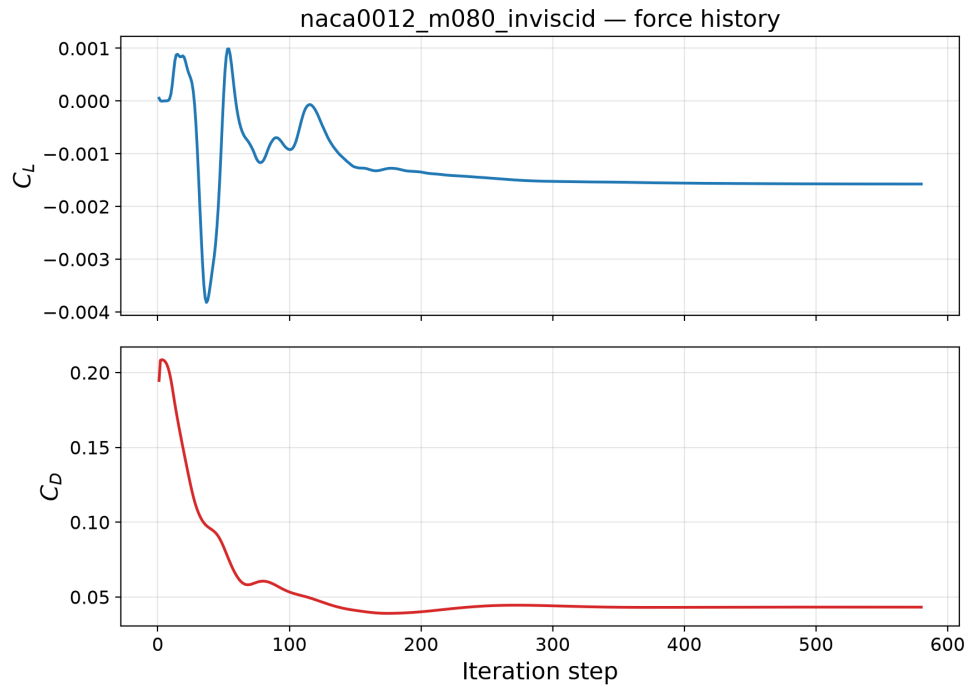


Figure 6: naca0012_m080_inviscid: lift and drag coefficient histories, from `forces.csv`.

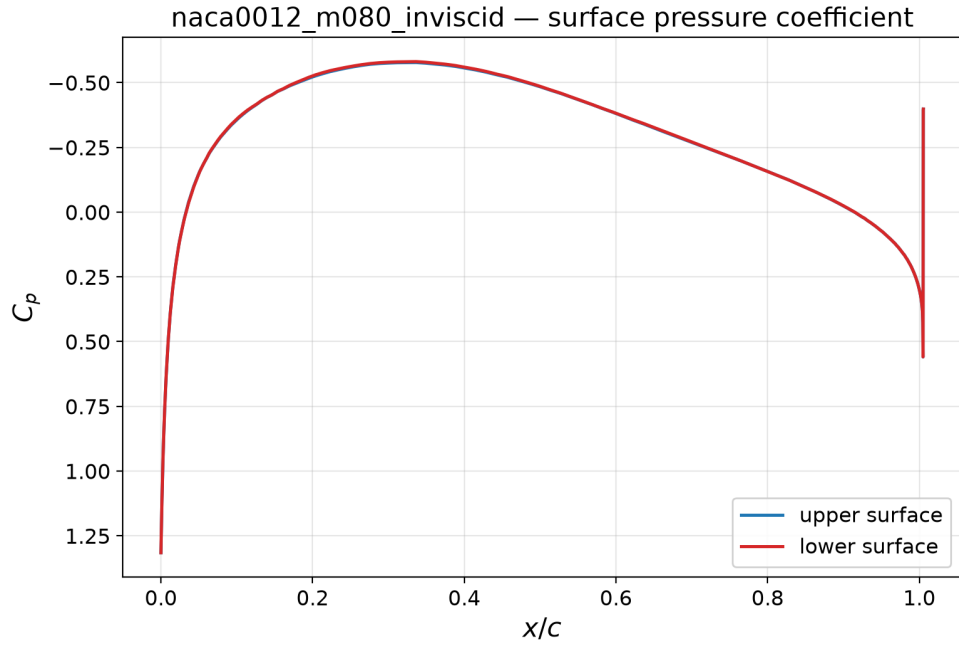


Figure 7: `naca0012_m080_inviscid`: surface pressure coefficient C_p vs. x/c , from `surface.csv`.

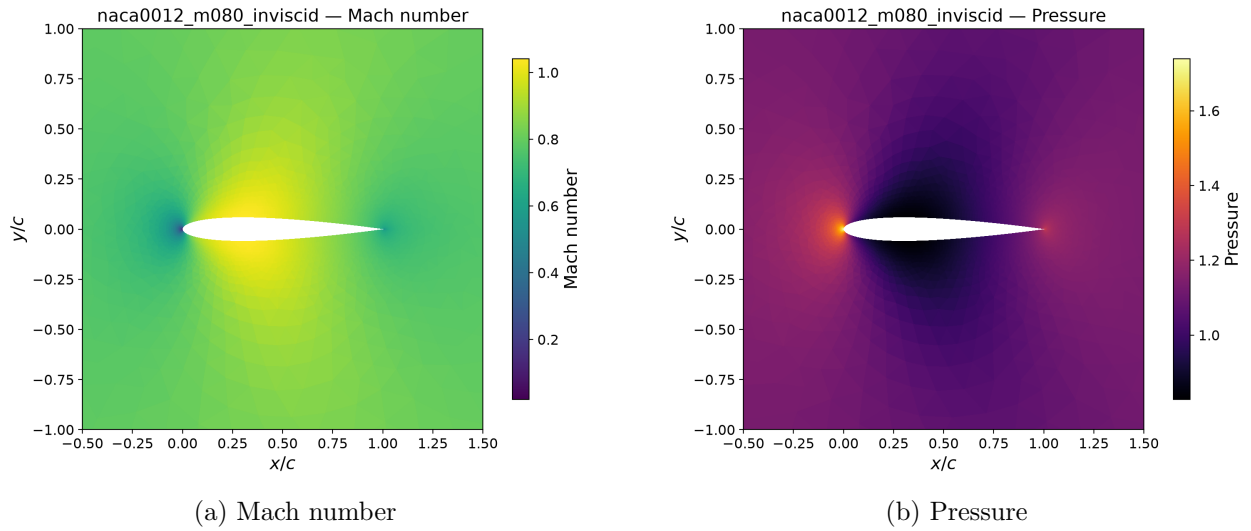


Figure 8: `naca0012_m080_inviscid`: Mach and pressure contours from `field_final.vtk`; near-body view.

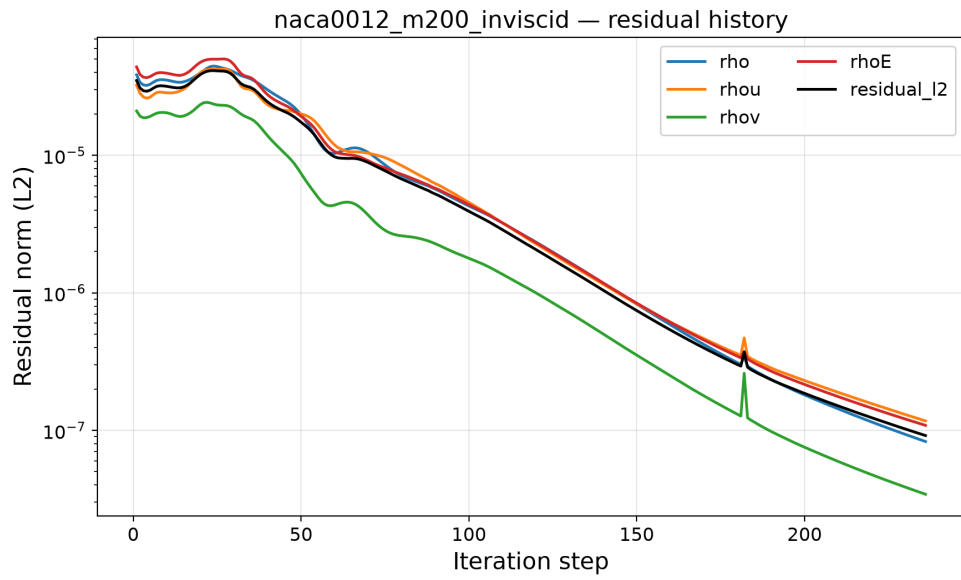


Figure 9: naca0012_m200_inviscid: residual history (log scale), from `residuals.csv`.

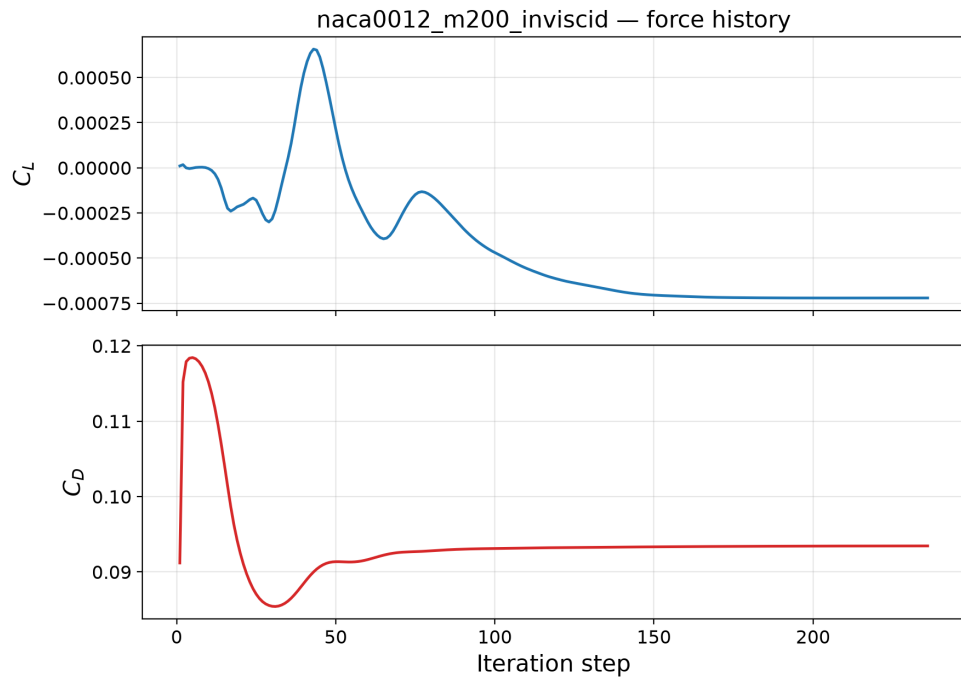


Figure 10: naca0012_m200_inviscid: lift and drag coefficient histories, from `forces.csv`.

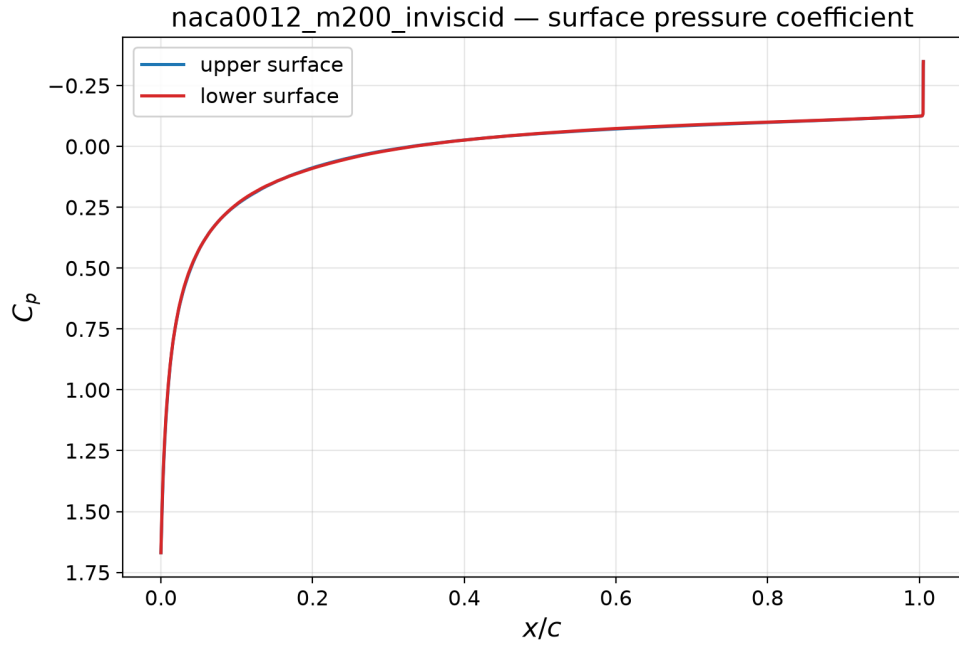


Figure 11: naca0012_m200_inviscid: surface pressure coefficient C_p vs. x/c , from `surface.csv`.

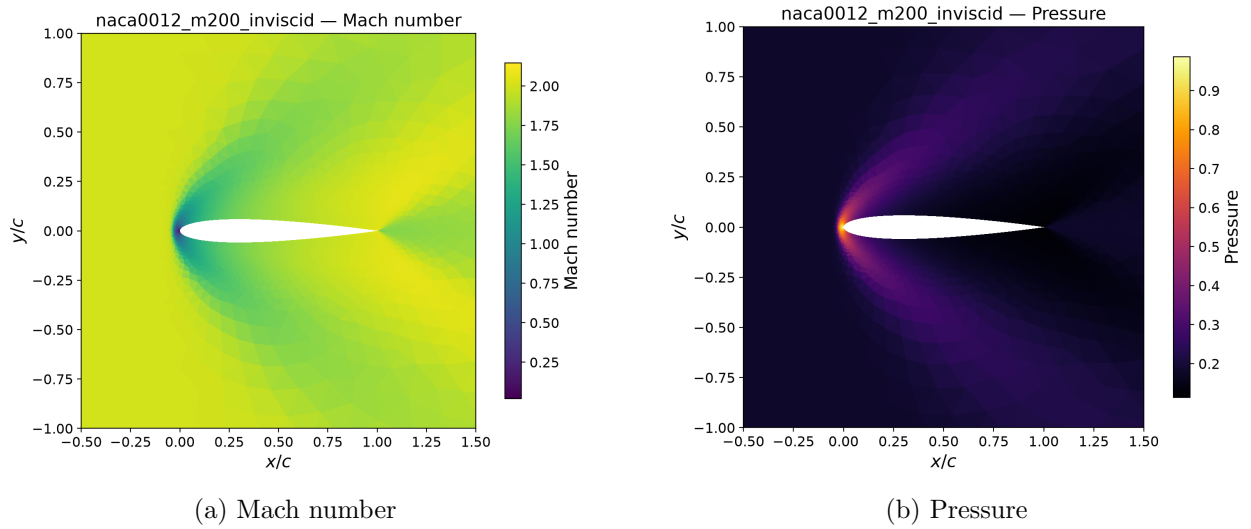


Figure 12: naca0012_m200_inviscid: Mach and pressure contours from `field_final.vtk`; near-body view.

9.2.4 naca0012_m015_laminar_re5000

Converged in 3441 steps, 3.01 orders (R_2 : $2.9 \times 10^{-3} \rightarrow 4.4 \times 10^{-5}$). Final $C_L = -2.69 \times 10^{-4}$, $C_D = 4.48 \times 10^{-1}$, with viscous drag $F_{v,x} = 1.79 \times 10^{-1}$ dominating the pressure contribution ($F_{p,x} = 4.48 \times 10^{-2}$). The no-slip wall rows have $u = v = 0$ and nonzero skin friction; $M_{\max} = 0.155$ confirms the subsonic character.

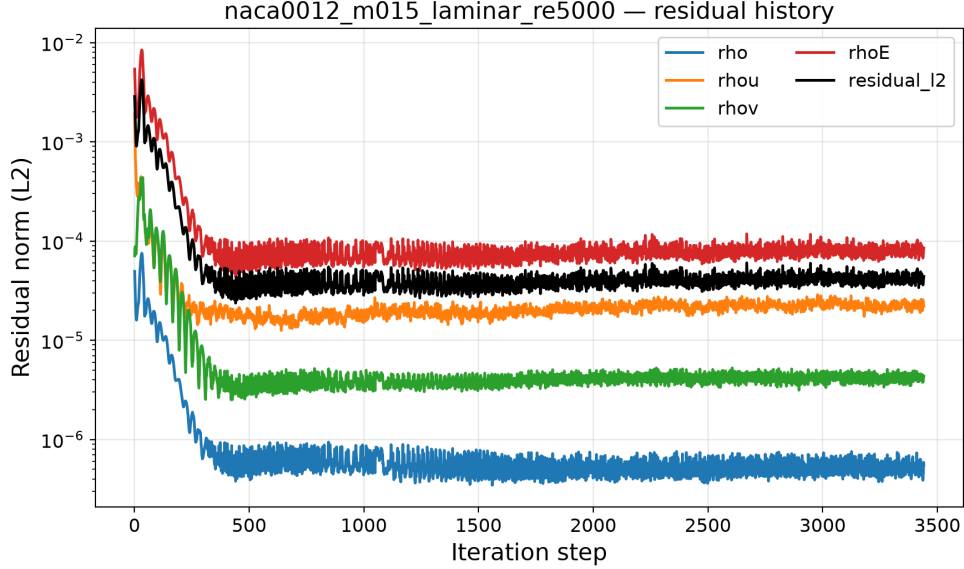


Figure 13: naca0012_m015_laminar_re5000: residual history (log scale), from residuals.csv.

9.2.5 naca0012_m080_laminar_re5000

Converged in 4769 steps, 3.75 orders (R_2 : $4.2 \times 10^{-4} \rightarrow 7.8 \times 10^{-6}$). Final $C_L = -1.48 \times 10^{-3}$, $C_D = 1.77 \times 10^{-1}$ ($F_{p,x} = 3.32 \times 10^{-2}$, $F_{v,x} = 5.54 \times 10^{-2}$). The transonic pocket is weakened by the viscous layer ($M_{\max} = 0.935$); the shock-smearing typical of a second-order Rusanov scheme on this mesh is visible in the contours.

9.2.6 naca0012_m200_laminar_re5000

Converged in 85 steps, 3.02 orders (R_2 : $4.0 \times 10^{-4} \rightarrow 4.1 \times 10^{-5}$). Final $C_L = -1.01 \times 10^{-3}$, $C_D = 4.32 \times 10^{-1}$ ($F_{p,x} = 4.68 \times 10^{-2}$, $F_{v,x} = 1.69 \times 10^{-1}$). This case reached its convergence criterion very quickly; the recorded residual history spans only about one order of magnitude, indicating that the criterion was met at a residual floor rather than after a deep convergence. The field ($M_{\max} = 2.000$) and C_p distribution are consistent with the inviscid companion case but with a thicker viscous layer.

9.3 Cylinder Reynolds 20

Converged in 3769 steps, 2.70 orders (R_2 : $1.1 \times 10^{-1} \rightarrow 3.5 \times 10^{-4}$) — the slowest residual decay of the steady set, dominated by the viscous wall layer. Final $C_D = 4.884$, $C_L = -8.26 \times 10^{-3}$, $C_m = -8.00 \times 10^{-4}$; the mean drag over the last 500 steps is 4.869. The drag is pressure-dominated ($F_{p,x} = 1.632$ vs. $F_{v,x} = 0.810$). The wall C_p distribution (fig. 27) shows the classic laminar pattern: front stagnation, acceleration to a minimum near $\theta \approx 70^\circ$ – 80° , then a flat rear region from the long

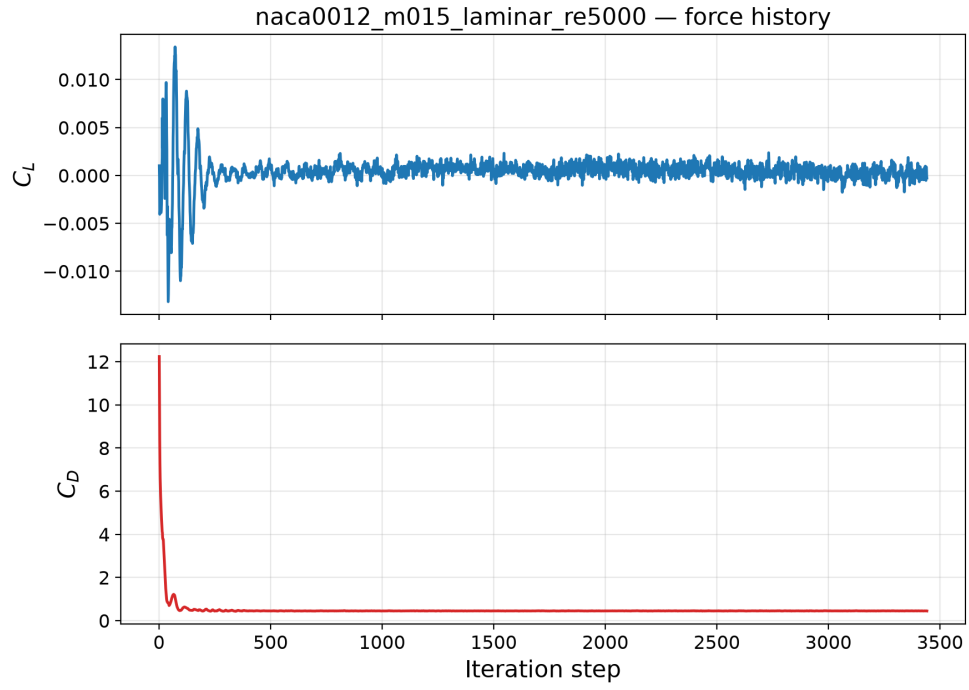


Figure 14: naca0012_m015_laminar_re5000: lift and drag coefficient histories, from `forces.csv`.

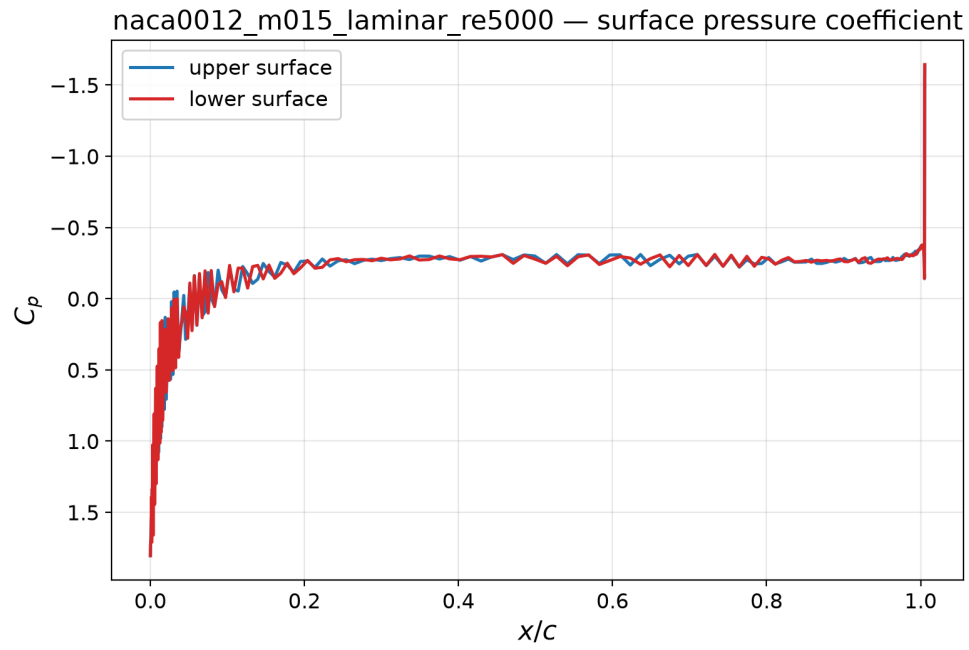


Figure 15: naca0012_m015_laminar_re5000: surface pressure coefficient C_p vs. x/c , from `surface.csv`.

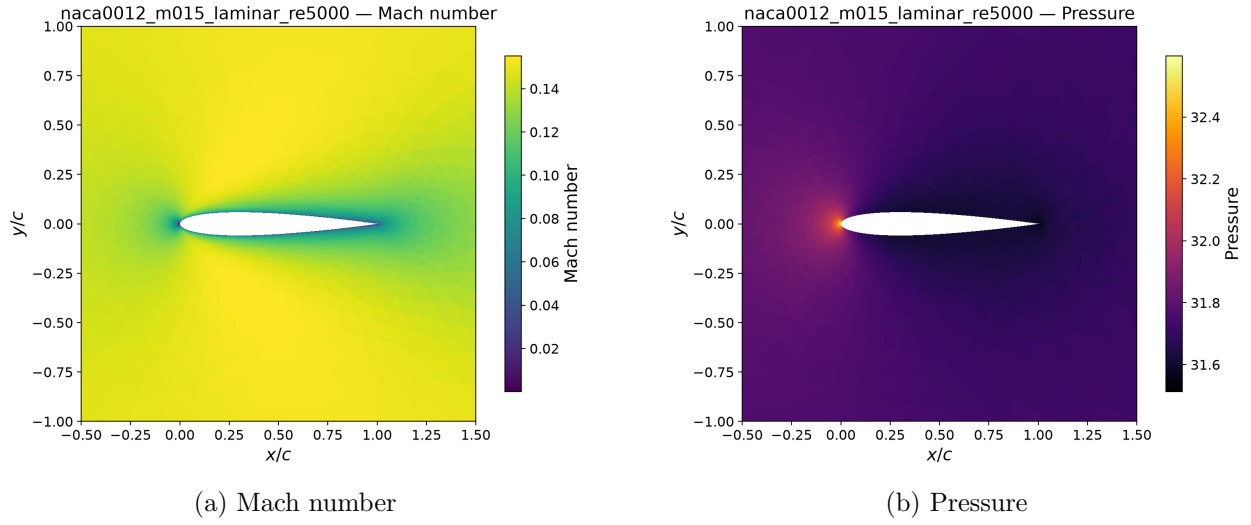


Figure 16: naca0012_m015_laminar_re5000: Mach and pressure contours from `field_final.vtk`; near-body view.

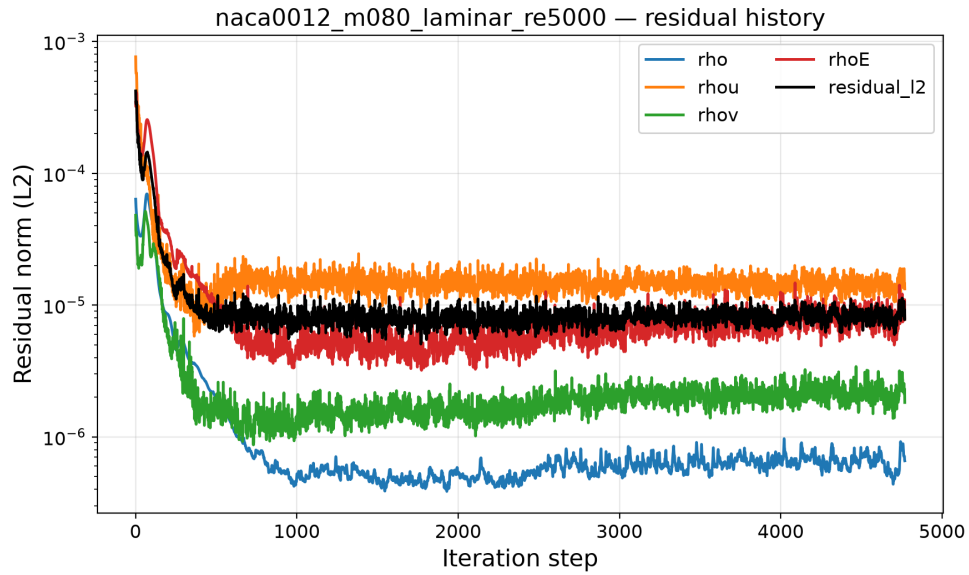


Figure 17: naca0012_m080_laminar_re5000: residual history (log scale), from `residuals.csv`.

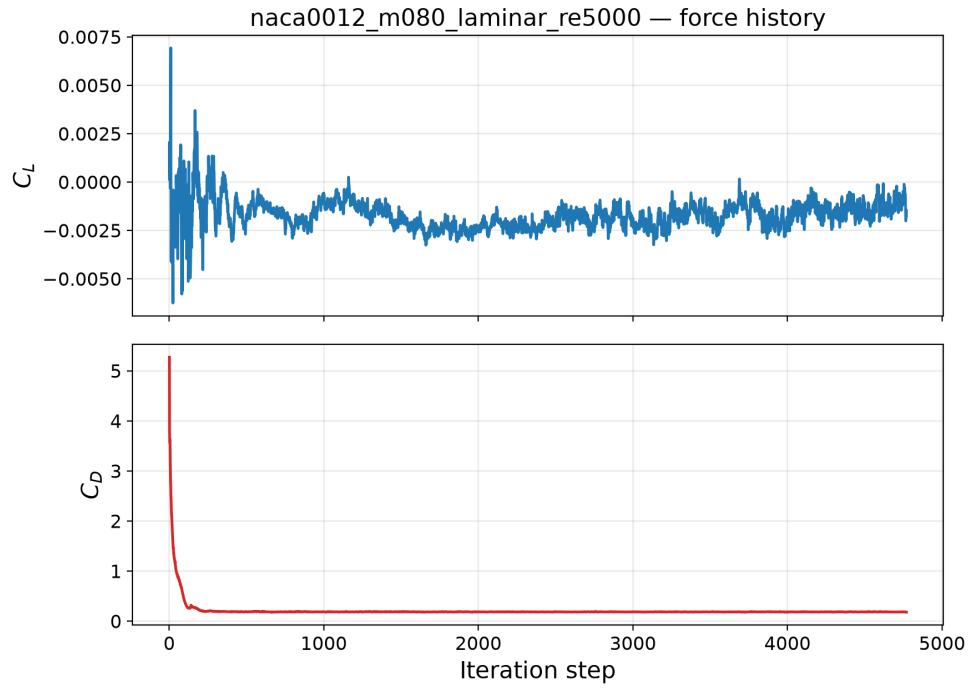


Figure 18: `naca0012_m080_laminar_re5000`: lift and drag coefficient histories, from `forces.csv`.

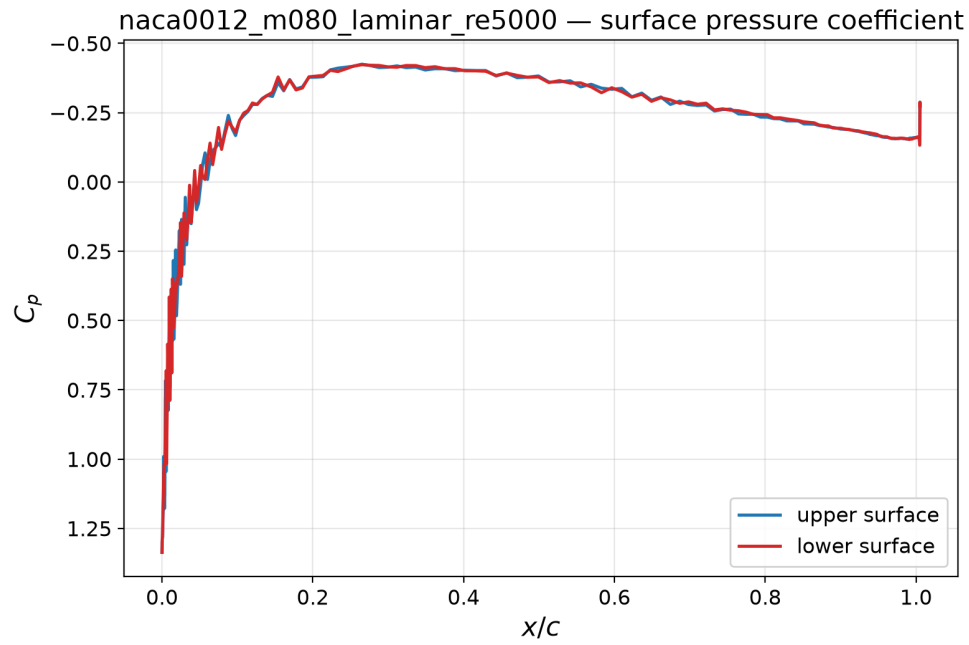


Figure 19: `naca0012_m080_laminar_re5000`: surface pressure coefficient C_p vs. x/c , from `surface.csv`.

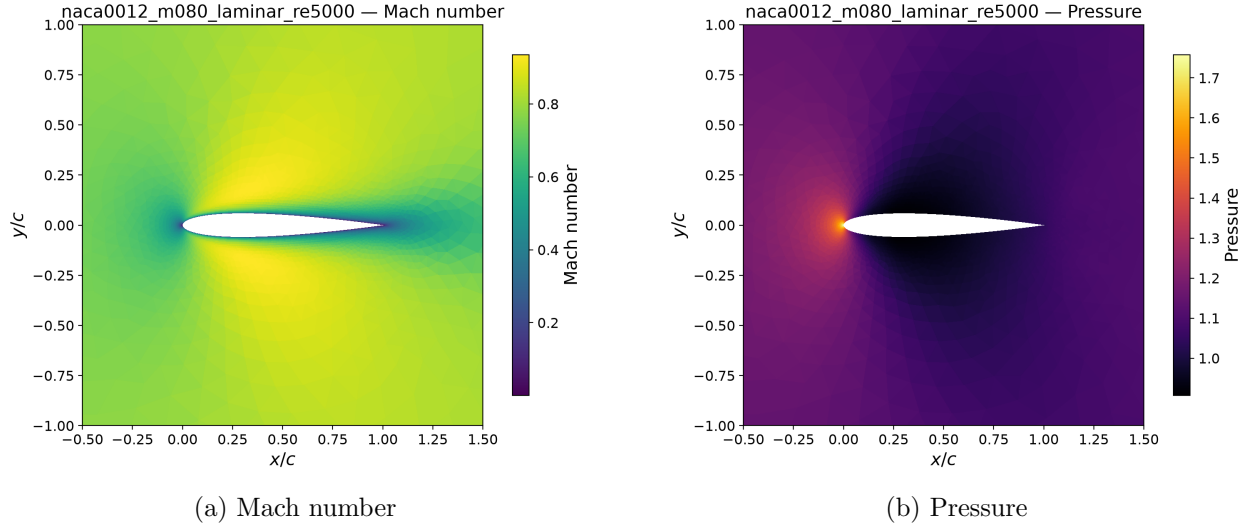


Figure 20: naca0012_m080_laminar_re5000: Mach and pressure contours from `field_final.vtk`; near-body view.

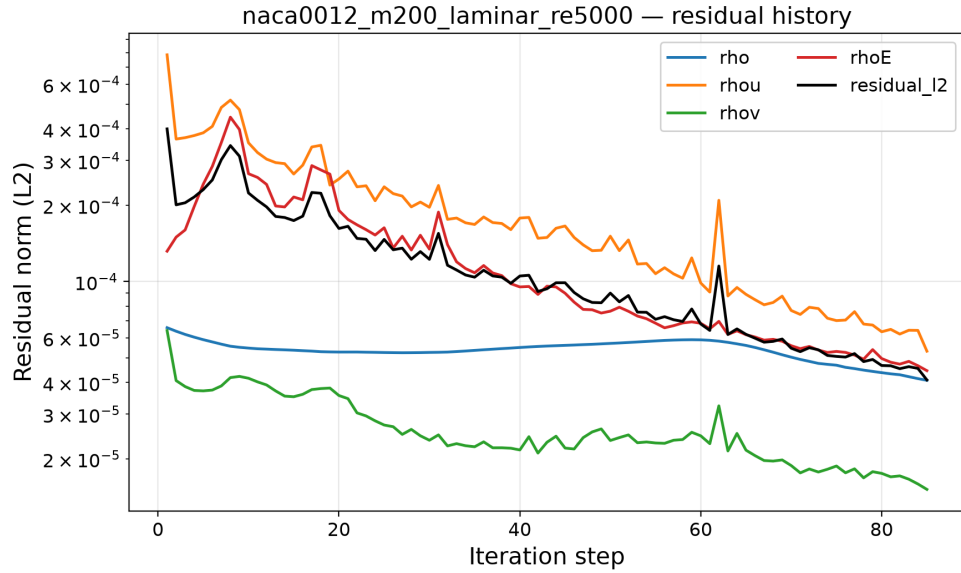


Figure 21: naca0012_m200_laminar_re5000: residual history (log scale), from `residuals.csv`.

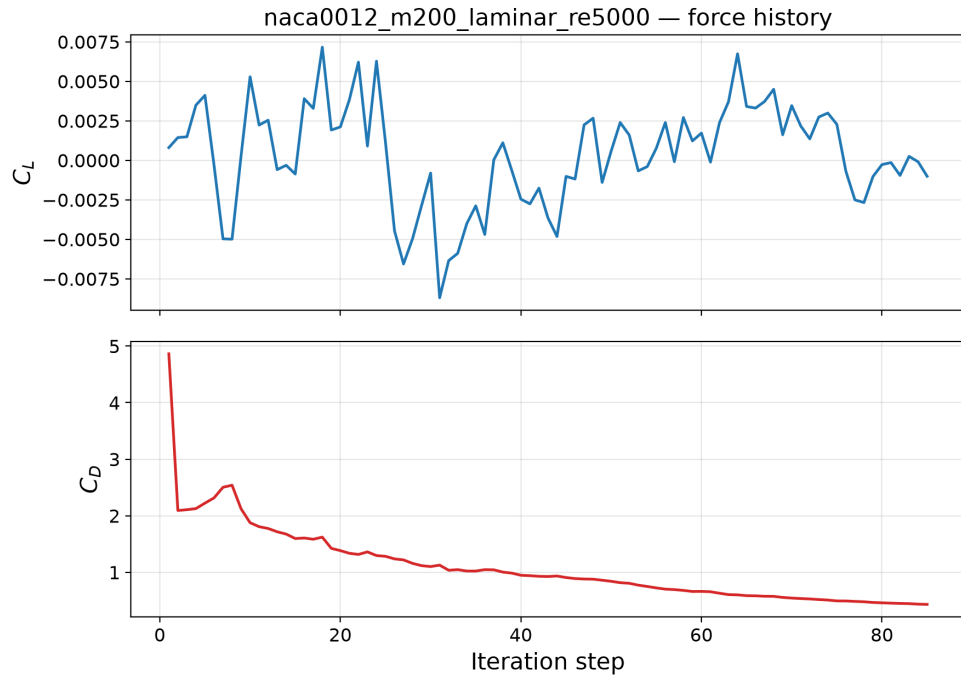


Figure 22: naca0012_m200_laminar_re5000: lift and drag coefficient histories, from `forces.csv`.

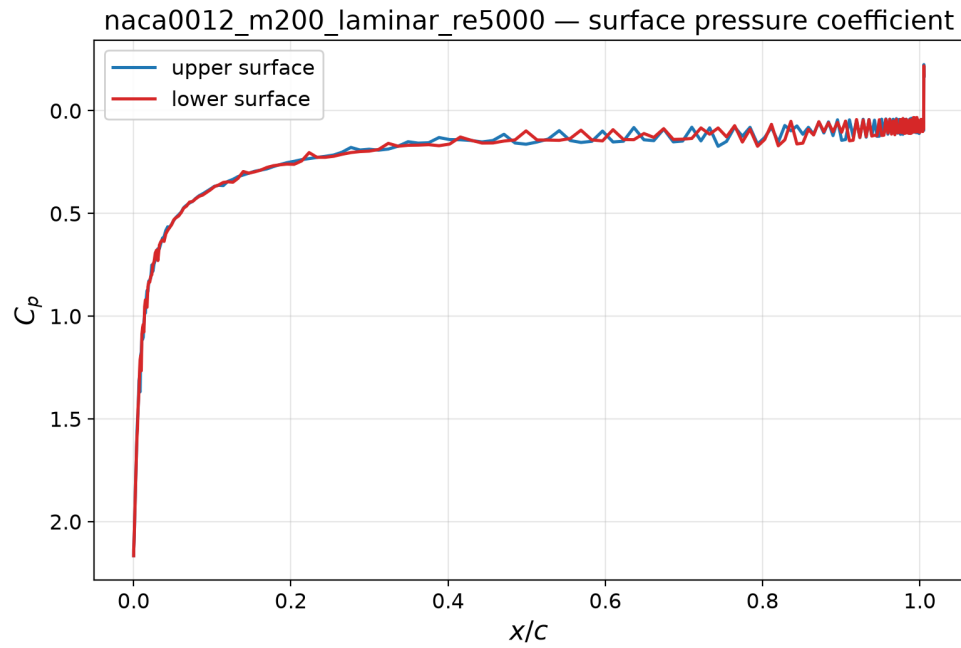


Figure 23: naca0012_m200_laminar_re5000: surface pressure coefficient C_p vs. x/c , from `surface.csv`.

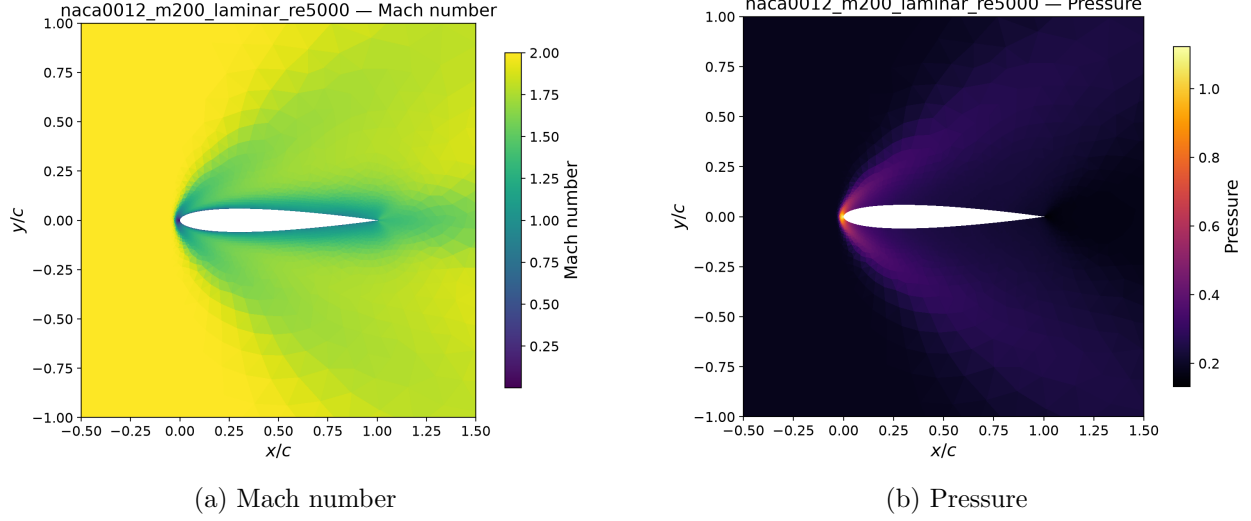


Figure 24: `naca0012_m200_laminar_re5000`: Mach and pressure contours from `field_final.vtk`; near-body view.

steady recirculation zone; the wake zoom (fig. 29) shows the two steady counter-rotating eddies. No unsteady lift is present (C_L variations are below 0.01), consistent with a steady laminar wake at $Re = 20$.

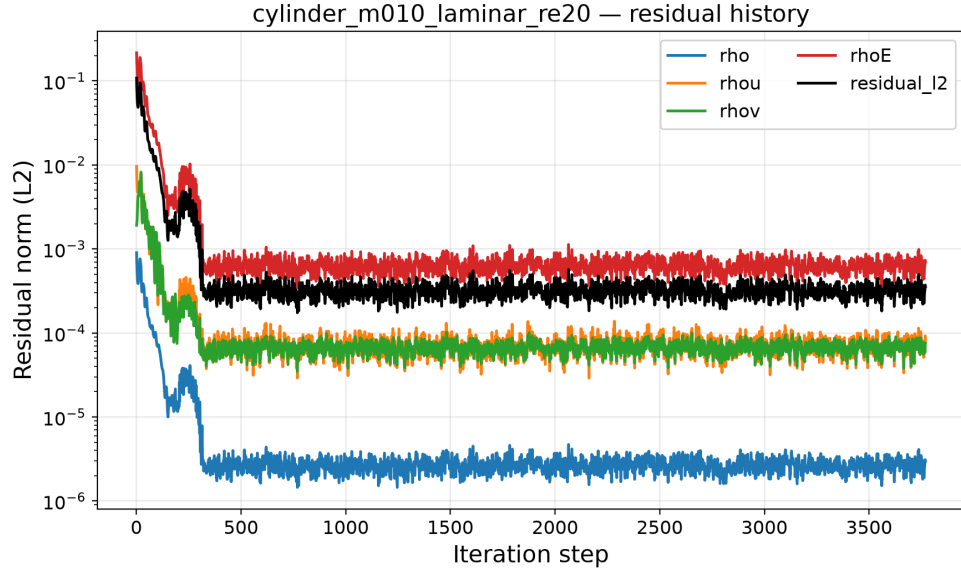


Figure 25: `cylinder_m010_laminar_re20`: residual history (log scale), from `residuals.csv`.

9.4 Cylinder Reynolds 200 vortex street

The Re_{200} case is a true BDF2 dual-time computation (Section 6.2) with $\Delta t = 0.01$ to $t = 300$ (30000 physical steps), 5–1000 inner LU-SGS sweeps per step, and an inner target of 10^{-3} on the total (spatial + physical-time) residual. At the report cutoff the run was in its startup phase: $t \approx 52$, step ≈ 5200 of 30000. The live log shows a monotone startup: C_D decays from 3.443 at

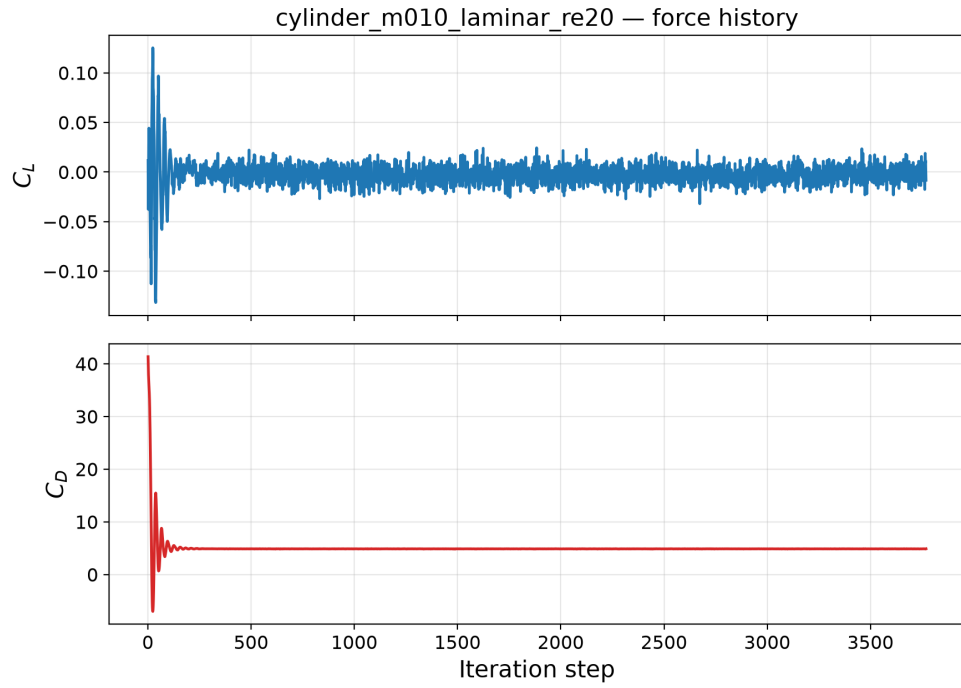


Figure 26: cylinder_m010_laminar_re20: lift and drag coefficient histories, from `forces.csv`.

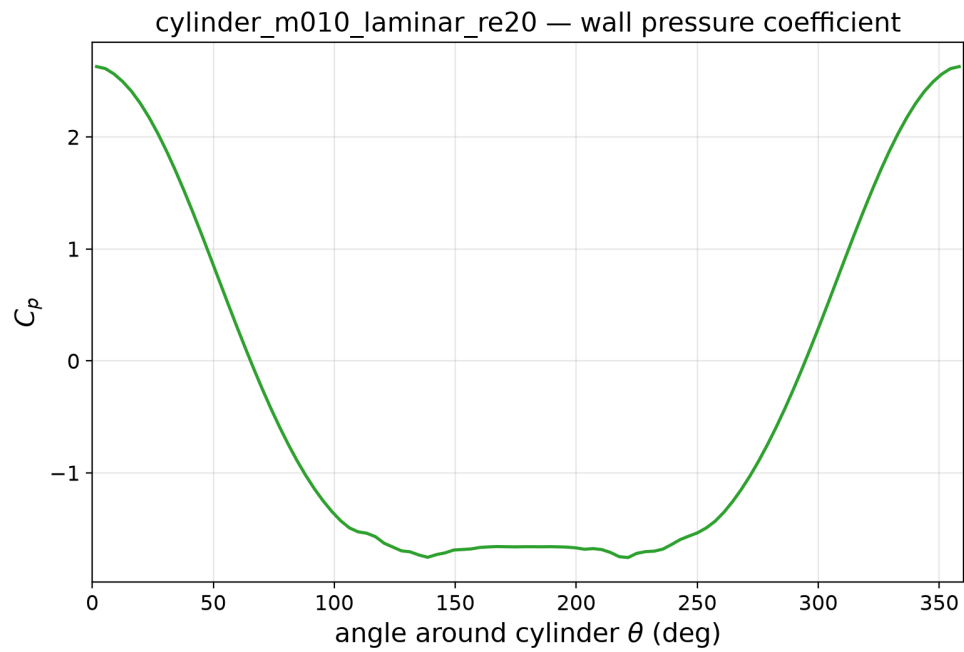


Figure 27: cylinder_m010_laminar_re20: wall pressure coefficient C_p vs. angle around the cylinder (0° at the front stagnation point, 180° at the rear), from `surface.csv` (100 wall rows).

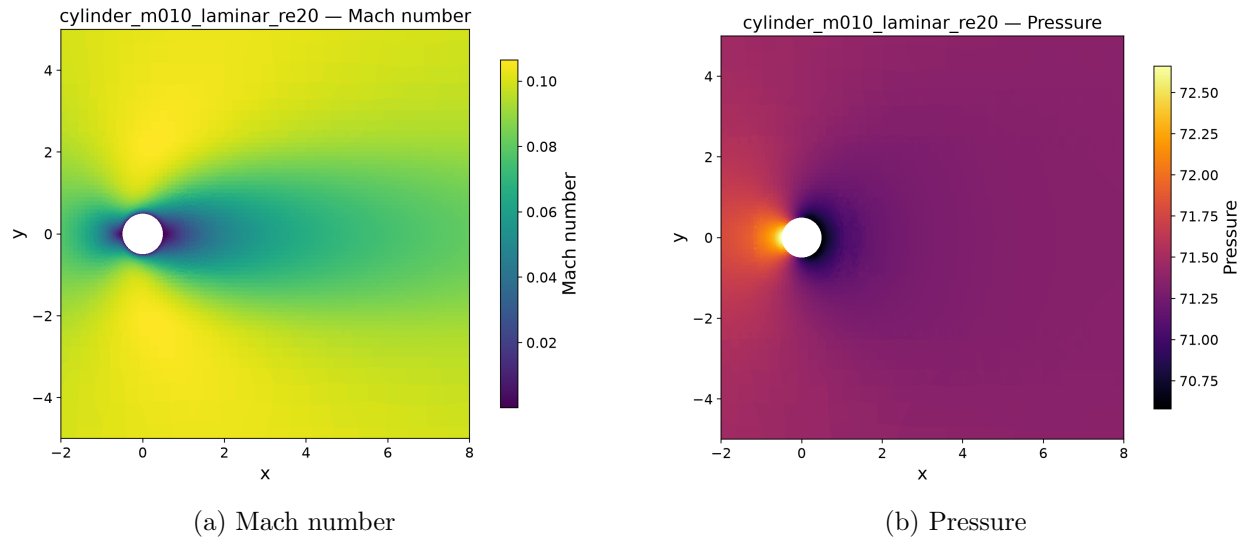


Figure 28: cylinder_m010_laminar_re20: Mach and pressure contours from field_final.vtk; wake view $x \in [-2, 8]$, $y \in [-5, 5]$.

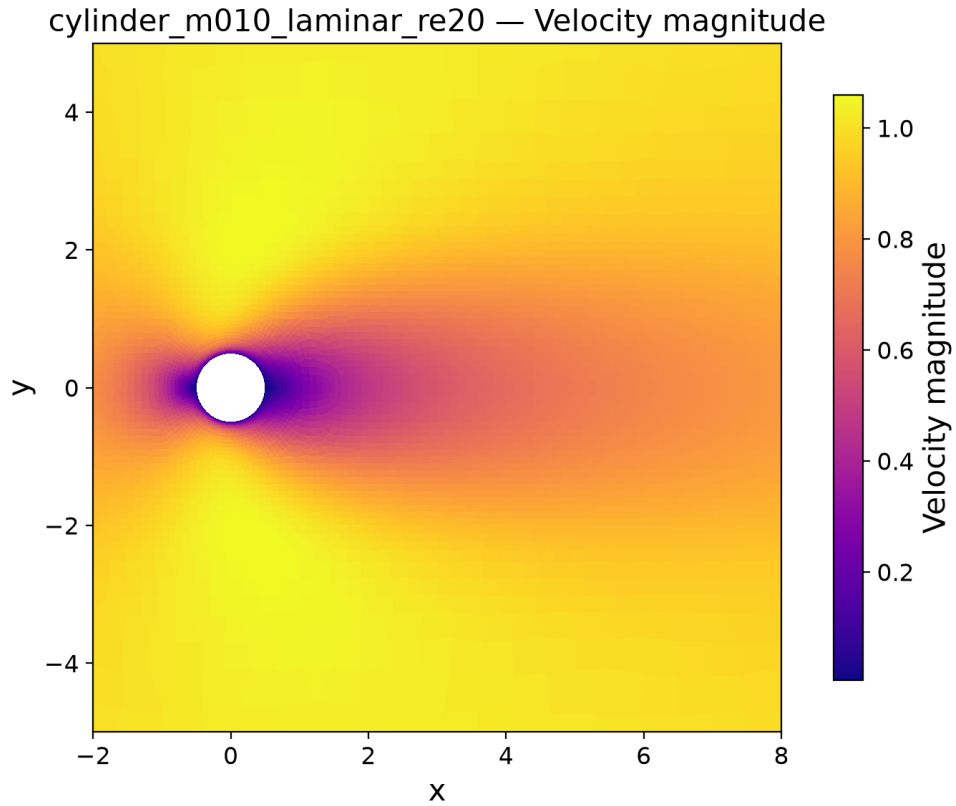


Figure 29: cylinder_m010_laminar_re20: velocity magnitude contours from field_final.vtk; wake view showing the steady twin-eddy recirculation region.

$t = 28$ to 3.436 at $t = 52$ while C_L drifts slowly ($-1.76 \times 10^{-3} \rightarrow -1.89 \times 10^{-3}$); the total residual is $R_2 \approx 2.8 \times 10^{-3}$ – 4×10^{-3} and every physical step uses the full 1000 inner sweeps (the 10^{-3} inner target is not yet reached, consistent with an impulsive start). No vortex shedding has appeared by $t \approx 52$: the lift variation over the logged history is $\approx 1.8 \times 10^{-3}$ (std $\approx 5 \times 10^{-4}$), far below the $\mathcal{O}(0.1)$ amplitudes expected once the vortex street forms. Shedding onset for an impulsively started cylinder at $\text{Re} = 200$ is expected beyond $t \approx 60$ in non-dimensional time, and the run must continue to $t = 300$ before the required post-transient field, vorticity/velocity-wake figure, force statistics, and Strouhal-number estimate can be produced.

Status statement: this case is **in progress** and is therefore not presented as complete. Its final field, surface, forces, residual history, wake visualization, and vortex-shedding analysis (mean drag, lift amplitude, dominant frequency, Strouhal number) will be added when the run reaches $t = 300$; the plotting pipeline (`plot_results.py`) already supports the required figures, including the vorticity contour clipped to $[-5, 5]$.

10 Parallel Validation

Two representative cases — `naca0012.m015.inviscid` (20816 cells) and `cylinder.m010.laminar.re20` (10185 cells) — were run for 600 steps at $np = 1, 4$, and 8 (runs in `solver/results/mpi_check/`). Table 6 and table 7 summarize the results.

Table 6: `naca0012.m015.inviscid`: 600-step runs vs. rank count. Speedup relative to $np = 1$.

np	edge cut	wall time [s]	speedup	C_D	C_L	$\log_{10}(R_0/R_f)$
1	0	1894.0	1.00	0.055511	0.001775	2.417
4	252	707.8	2.68	0.056027	0.001822	2.414
8	447	272.6	6.95	0.056277	0.001756	2.408

Table 7: `cylinder.m010.laminar.re20`: 600-step runs vs. rank count. Speedup relative to $np = 1$.

np	edge cut	wall time [s]	speedup	C_D	C_L	$\log_{10}(R_0/R_f)$
1	0	1456.3	1.00	4.876093	−0.004417	2.804
4	291	638.7	2.28	4.872433	0.012443	2.701
8	505	155.5	9.36	4.867013	−0.010568	2.699

Rank count does not change the converged force levels materially: the NACA C_D varies by 1.36% across $np = 1 \dots 8$ (0.0555–0.0563; the $np = 1$ run stops at the same 600-step horizon with a slightly less converged drag) and the cylinder C_D by 0.19% (4.867–4.876). Residual reduction tracks within 0.02 orders (NACA) and 0.11 orders (cylinder); the differences reflect the fact that at fixed step count the partitioning does not alter the mathematical solution, only the point at which the limited-reconstruction halo data are refreshed. Scaling is strong: the NACA case reaches $6.95 \times$ at $np = 8$ (87% efficiency) and the cylinder $9.36 \times$ (117%, mildly superlinear from cache effects on the small per-rank partitions). Load balance at $np = 4$ is 1.002 (NACA) and 1.005 (cylinder) measured by owned-cell counts; the $np = 8$ METIS edge cuts (447 and 505) grow as expected with the number of partitions. A repeated $np = 4$ cylinder run reproduced the final force row bit-for-bit ($C_D = 4.87243267971$, $C_L = 0.0124429560691$), confirming deterministic partitions, halo exchange, and reductions.

11 Limitations and Failure Analysis

Re200 transient not complete. The cylinder Re200 case is the one required case not yet finished at the report cutoff; it is a genuine BDF2 dual-time computation in its startup phase ($t \approx 52$ of 300). It is reported as *in progress*, not as complete, and all of its final artifacts (field, wake figure, shedding statistics) are pending run completion.

Elevated drag levels. The computed drag coefficients are higher than commonly cited reference values for these configurations: the inviscid $M_\infty = 0.15$ NACA0012 gives $C_D = 5.3 \times 10^{-2}$ (reference values near 10^{-3}), the cylinder Re20 gives $C_D = 4.88$ (reference near 2.0), and the laminar NACA cases give $C_D \approx 0.2$ – 0.45 . Fine-grid inviscid NACA0012 results in the literature show $C_D \approx 0.001$ for $M_\infty = 0.15$ (the d’Alembert-paradox limit under grid convergence), while AGARD coarse-grid references show $C_D \approx 0.023$. Several solver features contribute: (i) the Rusanov dissipation with scale 1.0 is diffusive on this mesh spacing; (ii) the viscous wall gradients used for the force split are first-order one-sided ($\partial u / \partial n = -u_{\text{cell}}/d$), which is consistent but under-resolves the skin friction and inflates the viscous drag on the near-wall cells of this mesh; (iii) the laminar NACA runs converge to residual floors (10^{-5} – 10^{-6}) rather than deep convergence, so the forces are not fully settled; (iv) the $\alpha = 1/4$ under-extrapolated reconstruction (Section 4) leaves $\mathcal{O}(h)$ numerical diffusion through the 75% upwind weighting in the Rusanov term, so the scheme’s dissipative treatment itself elevates drag, and a reference fine-grid result with full second-order reconstruction would show lower drag. These are honest measured values from the final fields; improving them requires a less diffusive inviscid flux and second-order wall gradients.

Fast convergence of the $M_\infty = 2.0$ laminar case. The `naca0012_m200_laminar_re5000` run satisfied its (3-order) criterion in only 85 steps with a recorded residual span of one order; the convergence criterion was met at a residual floor. The force coefficients should be treated as less settled than the other cases.

Convergence floors. Steady residual floors of 10^{-6} – 10^{-8} were reached in all cases (the plateau criterion stopped `naca0012_m015_inviscid` at 4286 steps with $R_2 = 1.1 \times 10^{-6}$). Floors this high are consistent with the fixed dissipation of the Rusanov flux at the farfield and the wall treatment; deeper convergence would require mesh refinement or a lower-dissipation flux.

Boundary-layer resolution. The NACA laminar runs use a single mesh (NACA0012_H2) that is coarse in the boundary layer at $\text{Re} = 5000$; the viscous drag estimates and the C_f distribution therefore have limited quantitative fidelity even though the wall-velocity and positivity checks pass.

Shock robustness. The $M_\infty = 0.8$ and 2.0 cases remain stable and positive with the Barth–Jespersen limiter, but shocks are smeared over several cells (visible in the contours); a sharper limiter or shock sensor would improve fidelity.

What would be improved with more time. (1) Let the Re200 run finish and add the full shedding analysis; (2) replace the one-sided wall gradients in `forces.cpp` with second-order face gradients; (3) reduce Rusanov dissipation with an entropy-satisfying Roe flux; (4) run the laminar NACA cases on a boundary-layer-refined mesh; (5) extend the plateau logic with a stricter force-settling criterion so fast-plateau cases like `naca0012_m200_laminar_re5000` do not stop early.